

PROJECT TITLE:

**SECURITY OPERATIONS CENTER
(SOC) MONITORING SYSTEM**

Submitted By

Name: Tushar Gauswami

Course: Cyber Security

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty members, project guide, and institution for providing me with the opportunity to complete this project on the **Security Operations Center (SOC) Monitoring System**.

This project gave me practical exposure to real-world cybersecurity operations such as security monitoring, event log analysis, SIEM implementation, alert generation, dashboard creation, and network traffic investigation. Through this project, I gained hands-on experience with Splunk Enterprise, Windows Event Viewer, and Wireshark.

The knowledge gained through this work has improved my understanding of modern threat detection methodologies and SOC operations.

EXECUTIVE SUMMARY

A Security Operations Center (SOC) is the foundation of organizational cybersecurity defense. It continuously monitors systems, analyzes events, detects suspicious activities, and responds to security incidents.

With increasing cyber threats such as brute-force attacks, malware infections, phishing campaigns, privilege escalation, and unauthorized access attempts, organizations require continuous monitoring to maintain security posture. SOC analysts perform this responsibility using SIEM tools that centralize logs from multiple systems.

In this project, a SOC Monitoring System was implemented using Splunk Enterprise. Windows security logs were collected from Windows Event Viewer and imported into Splunk for analysis. Different SPL (Search Processing Language) queries were used to analyze authentication activities.

The project includes:

- Security log collection
- Log ingestion into Splunk
- Failed login detection
- Successful login tracking
- Dashboard creation
- Alert generation
- Incident investigation
- DNS traffic analysis using Wireshark

The SOC dashboard provided visualized data of login events and system activities. Alerts were configured to detect suspicious failed logins. Wireshark analysis provided visibility into DNS communications.

This project demonstrates the practical implementation of a SOC environment for threat detection and incident response.

TABLE OF CONTENTS

1. Introduction
2. Objectives
3. Scope
4. Methodology
5. Tools Used
6. Windows Log Collection
7. Splunk Data Ingestion
8. SPL Query Analysis
9. Suspicious Activity Analysis
- 10.SOC Dashboard Creation
- 11.Alert Configuration
- 12.Incident Investigation
- 13.Wireshark Traffic Analysis
- 14.Threat Analysis Report
- 15.Risk Assessment
- 16.Conclusion
- 17.Recommendations
- 18.References
- 19.Appendix

INTRODUCTION

What is SOC?

A Security Operations Center (SOC) is a centralized operational unit responsible for monitoring, detecting, analyzing, and responding to cybersecurity incidents.

It acts as the command center of an organization's security infrastructure.

The main purpose of a SOC is to ensure:

- Continuous security monitoring
- Early threat detection
- Incident response
- Risk reduction
- Compliance monitoring

A SOC operates 24×7 to ensure that any suspicious activity is detected immediately.

Importance of SOC

In modern organizations, cyber threats occur constantly. Attackers attempt to gain unauthorized access using multiple methods such as:

- Brute-force attacks
- Password spraying
- Malware deployment
- Phishing
- DNS abuse
- Privilege escalation

Without SOC monitoring, these threats may remain undetected.

The SOC provides:

Function	Benefit
Monitoring	Continuous visibility
Detection	Early identification
Analysis	Threat understanding
Response	Faster mitigation
Reporting	Better compliance

Role of SOC Analysts

SOC analysts are cybersecurity professionals responsible for:

Level 1 Analyst (Monitoring)

They monitor logs and identify suspicious events.

Level 2 Analyst (Investigation)

They investigate incidents and correlate events.

Level 3 Analyst (Response)

They perform advanced threat hunting and remediation.

In this project, the role of SOC analysts was simulated using Splunk.

Importance of SIEM in SOC

SIEM (Security Information and Event Management) collects logs from multiple devices and centralizes them.

Benefits of SIEM:

- Log aggregation
- Event correlation
- Dashboard visualization
- Alerting
- Threat detection

In this project, Splunk Enterprise was used as SIEM.

OBJECTIVES

The objectives of this project are:

1. To collect Windows Security logs.
2. To import logs into Splunk.
3. To analyze authentication events.
4. To detect failed login attempts.
5. To monitor successful logins.
6. To create SOC dashboards.
7. To configure alerts.
8. To investigate incidents.
9. To analyze network traffic.

SCOPE

This project focuses on:

- Host-based security monitoring
- Authentication event analysis
- SIEM-based centralized monitoring
- Security event visualization
- Alert-based threat detection
- DNS traffic monitoring

The project is limited to a Windows host environment.

METHODOLOGY

The methodology followed:

Step 1: Log Collection

Windows logs collected from Event Viewer.

Step 2: Log Import

Logs imported into Splunk.

Step 3: Event Analysis

Security events filtered and analyzed.

Step 4: Dashboard Creation

Visual dashboards built.

Step 5: Alert Configuration

Alerts configured for failed logins.

Step 6: Network Analysis

Wireshark used for DNS analysis.

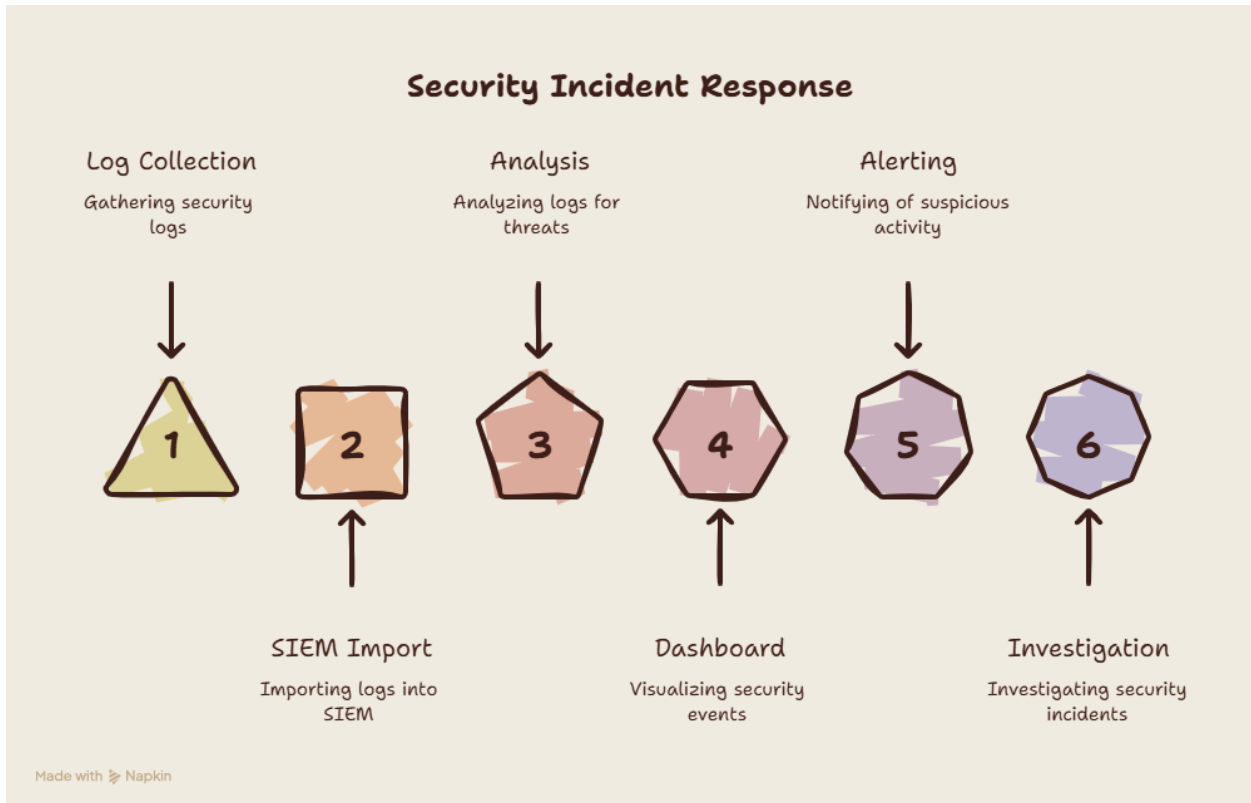
Step 7: Incident Investigation

Suspicious activities investigated.

Flow:

Log Collection → SIEM Import → Analysis → Dashboard → Alerting → Investigation

Figure 1.1: SOC Methodology



TOOLS USED

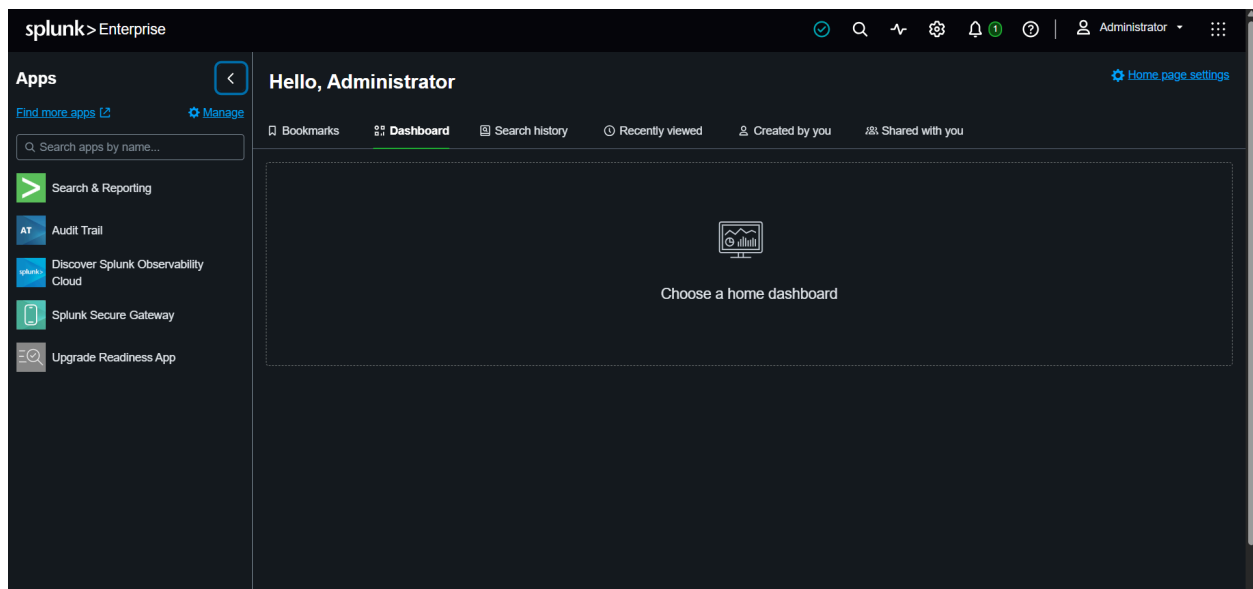
1. Splunk Enterprise

Splunk is a SIEM platform used for:

- Log collection
- Event analysis
- Dashboard creation
- Alert configuration

It uses SPL queries for searching and analyzing logs.

Figure 1.2: Tools Used



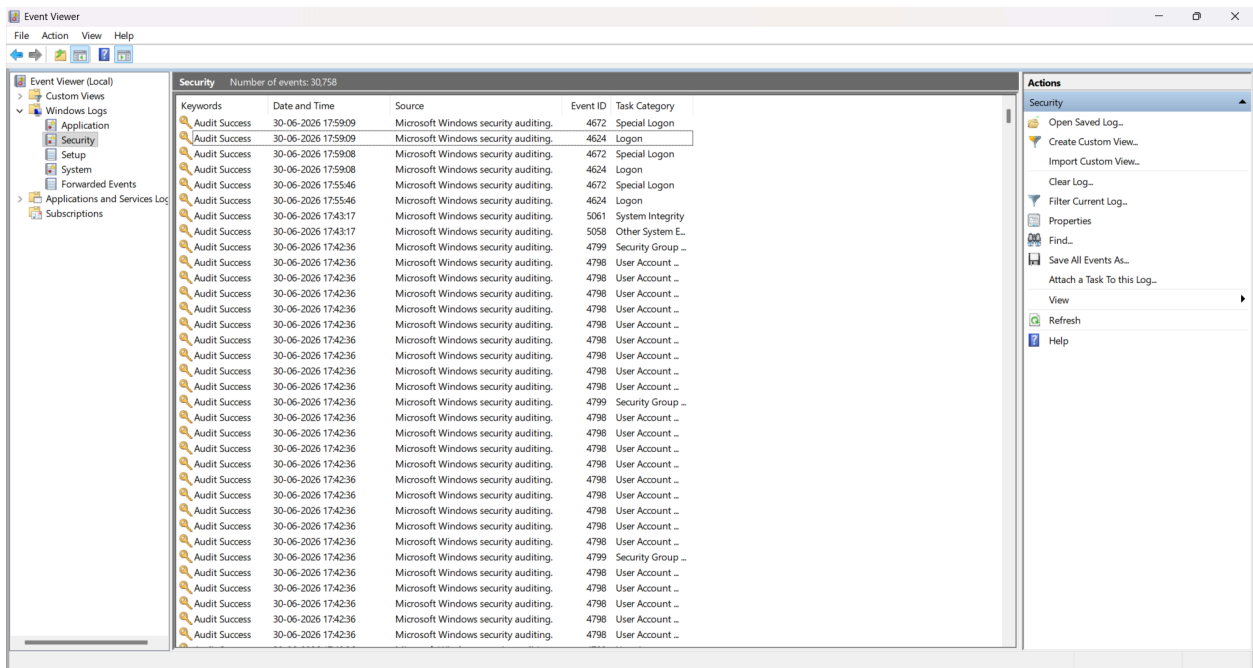
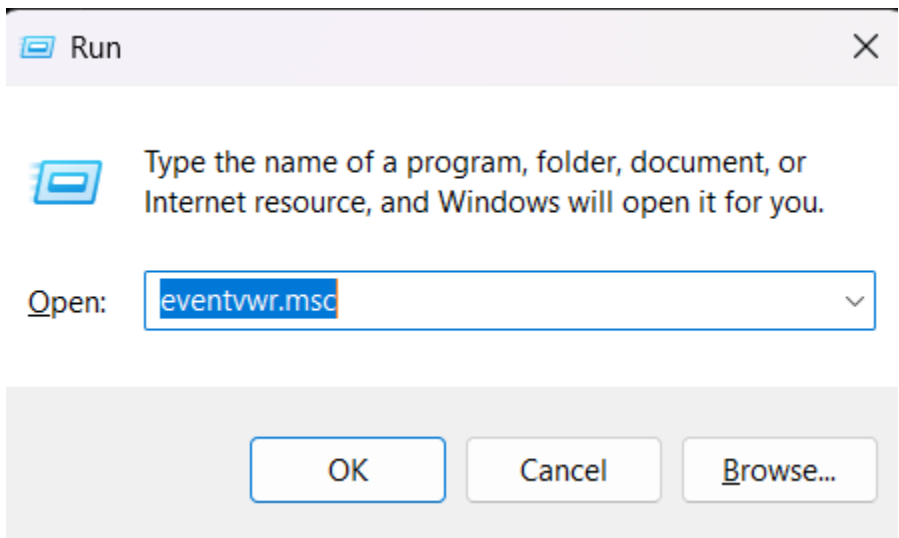
2. Windows Event Viewer

Event Viewer is used to collect Windows logs.

Logs analyzed:

- Security logs
- Authentication logs
- System logs

Figure 1.3: Tools Used

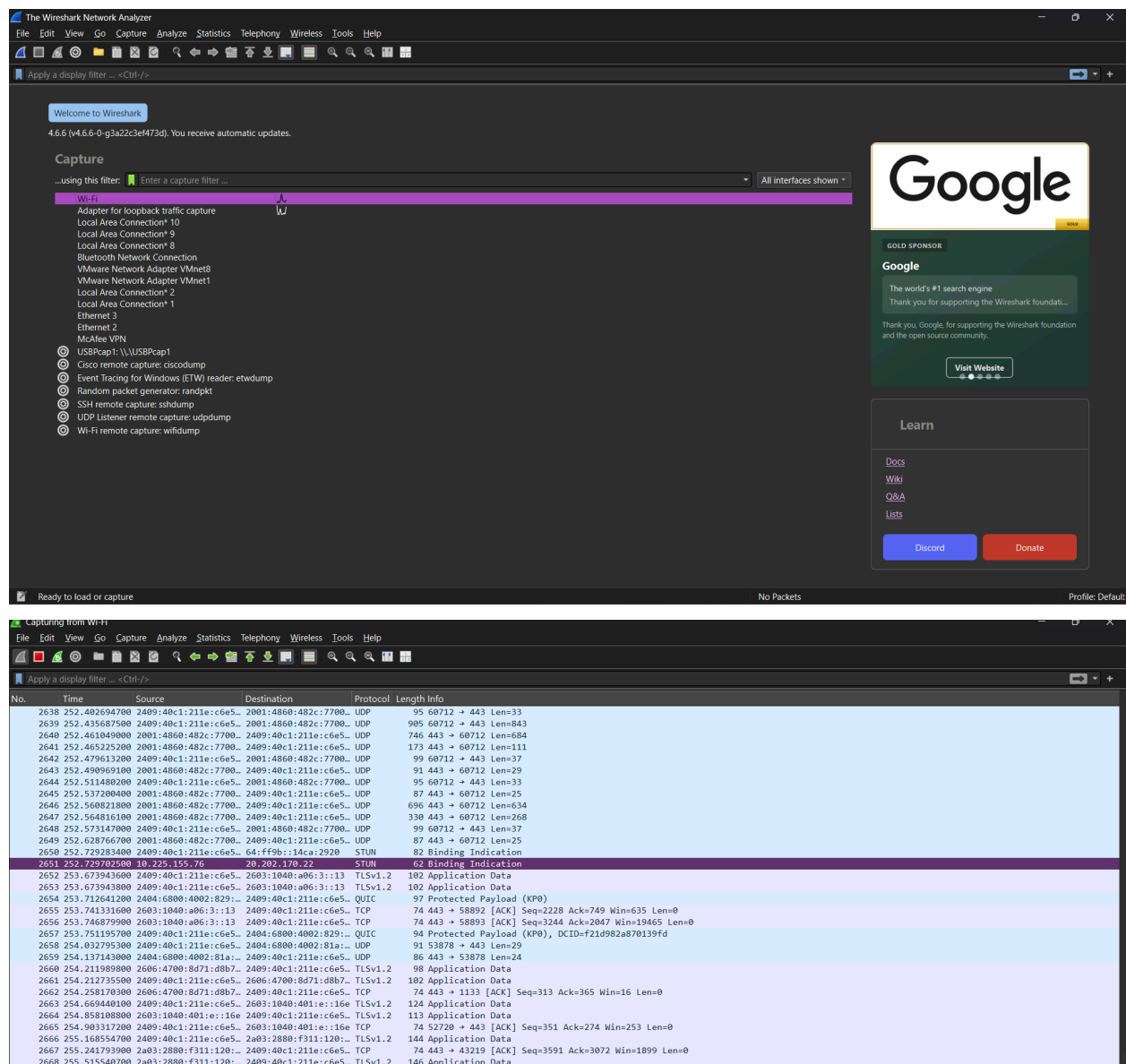


3. Wireshark

Wireshark captures packets and helps analyze:

- DNS traffic
- TCP connections
- HTTP traffic
- UDP traffic

Figure 1.4: Tools Used



PART 2: WINDOWS LOG COLLECTION

Windows Security logs were collected from:

Windows Logs → Security

Total events recorded:

30,900

These logs contain important information regarding user authentication, privilege changes, and security events.

Important Event IDs:

Event ID	Description
4624	Successful login
4625	Failed login
4634	Logout
4672	Special privileges
4688	Process creation

Event ID 4625 Analysis

From your screenshot:

Field	Value
Event ID	4625
Log Type	Security
Level	Information
Keywords	Audit Failure

Time	29-06-2026 16:11:03
Host	TUSHAR

Event ID 4625 records failed login attempts.

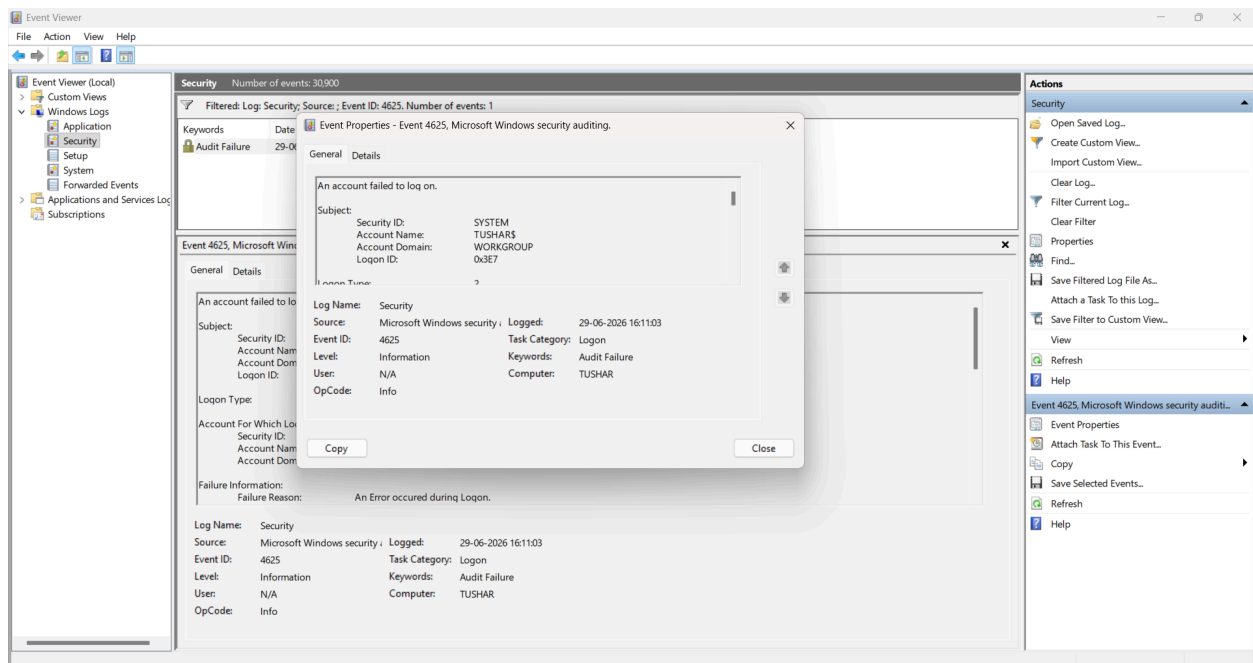
This event is important because attackers often perform repeated failed logins during brute-force attacks.

The “Audit Failure” status indicates unsuccessful authentication.

SOC analysts use this event to detect:

- Brute-force attacks
- Unauthorized access
- Account misuse

Figure 2.1: Event ID 4625



PART 3: SPLUNK DATA INGESTION

Introduction to Log Ingestion

Log ingestion is one of the most important processes in any Security Information and Event Management (SIEM) system. It is the process of collecting logs from different sources and importing them into a centralized platform for analysis.

In this project, Windows Security logs were imported into Splunk Enterprise for centralized monitoring.

The purpose of importing logs into Splunk is:

- To centralize all security events.
- To make searching easier.
- To detect suspicious activities quickly.
- To correlate events.
- To create dashboards and alerts.

Without log ingestion, analysis would not be possible.

Why Splunk for Log Ingestion?

Splunk is widely used in SOC environments because it provides:

Feature	Benefit
Real-time log collection	Faster monitoring
Centralized storage	Easy analysis
SPL queries	Powerful searching
Dashboards	Better visualization
Alerts	Automatic detection

Splunk allows logs from Windows, Linux, firewalls, IDS/IPS, and applications to be imported into one system.

Steps Performed for Log Ingestion

The following steps were performed in Splunk:

Step 1: Open Splunk Enterprise

The Splunk dashboard was opened in the web browser.

This acts as the central SIEM console.

Step 2: Navigate to Add Data

The **Add Data** feature was selected.

Purpose:

To start the process of importing logs.

Step 3: Select Data Source

The source selected was:

Windows Event Logs

This source directly collects Windows Security events.

From the screenshot:

Parameter	Value
Source Type	Windows Event Logs
Event Log	Security
Host	TUSHAR
Index	main

Step 4: Configure Host

Host selected:

TUSHAR

The host field identifies the machine generating logs.

This is important because SOC analysts often investigate events by host.

Step 5: Configure Index

Index used:

main

In Splunk, an index is a storage location for logs.

Why indexing matters:

- Faster searching
- Better organization
- Efficient retrieval

Step 6: Review Configuration

The review page confirms:

- Event source
- Host
- Index
- Input settings

This prevents configuration mistakes.

Step 7: Submit

After reviewing, logs were successfully ingested into Splunk.

This made them available for SPL analysis.

Screenshot Analysis

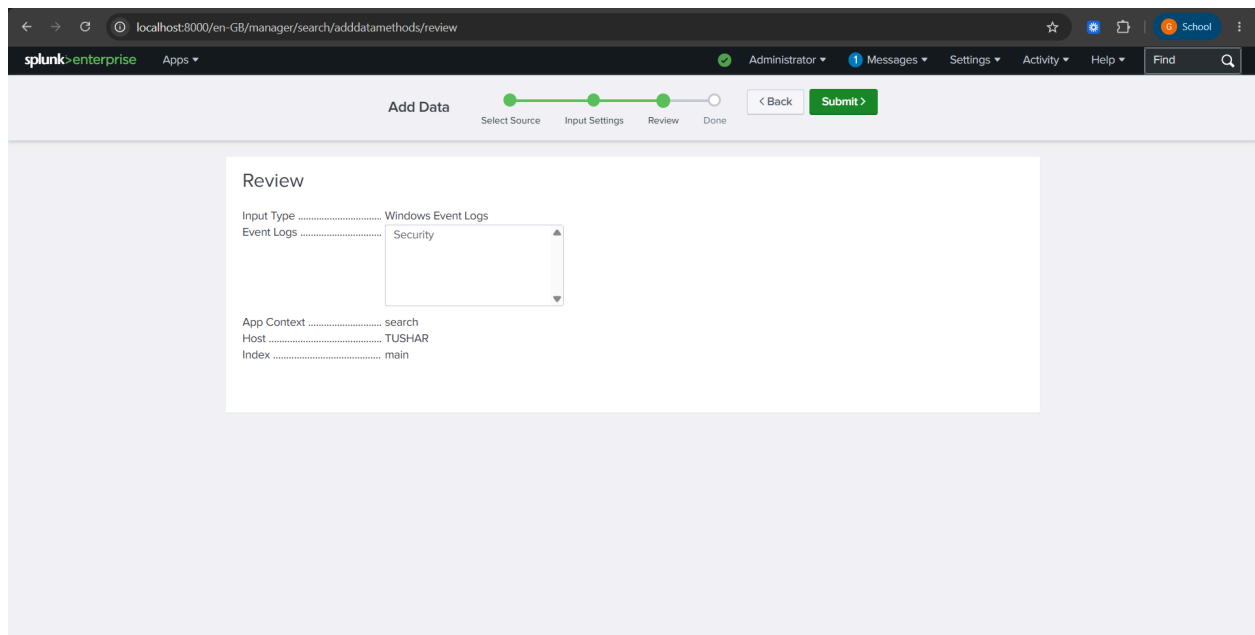
From your Splunk review screenshot:

Observed values:

Field	Observation
Input Type	Windows Event Logs
Log Type	Security
Host	TUSHAR
Index	main

This confirms the successful connection between Windows logs and Splunk.

Figure 3.1: Splunk Data Review



Importance of Centralized Logging

Centralized logging provides:

Better Visibility

All logs are in one place.

Faster Detection

SOC analysts can detect threats quickly.

Easier Investigation

Attack timelines can be reconstructed.

Correlation

Different events can be connected.

Example:

A failed login followed by a privilege escalation can indicate compromise.

Log Verification

After ingestion, logs were verified.

Verification query:

index=main

Purpose:

This query checks whether logs were successfully imported.

If events appear, ingestion is successful.

Verification Findings

The following was observed:

Parameter	Result
Logs Imported	Successful
Host Visibility	Available
Event Searchable	Yes
Ready for Analysis	Yes

This confirms that the SIEM environment is operational.

PART 4: SPL QUERY ANALYSIS

Introduction to SPL

SPL (Search Processing Language) is Splunk's query language used for searching and analyzing logs.

It helps SOC analysts:

- Search logs
- Filter events
- Aggregate data
- Detect attacks
- Generate reports

SPL is powerful because it can analyze millions of logs quickly.

Failed Login Detection

Query used:

index=main EventCode=4625

Explanation of Query

Breakdown:

Part	Meaning
index=main	Search logs from main index
EventCode=4625	Filter failed login events

This query specifically searches for failed authentication attempts.

This is critical in detecting:

- Brute-force attacks
- Password spraying
- Unauthorized access attempts

Security Importance of EventCode 4625

EventCode 4625 indicates login failure.

Reasons can include:

- Wrong password
- Account disabled
- Invalid username
- Policy restrictions

Repeated occurrences may indicate malicious activity.

Practical Findings

From Event Viewer:

Field	Value
Event ID	4625
Type	Audit Failure
Host	TUSHAR
Time	29-06-2026 16:11:03

This confirms that failed authentication events exist.

These events can be used for brute-force detection.

Successful Login Detection

Query used:

index=main EventCode=4624

Query Explanation

EventCode 4624 indicates successful authentication.

This event helps analysts identify:

- Legitimate user access
- Unusual login timing
- Suspicious successful logins after failures

This is useful for attack correlation.

Example:

Multiple failed logins followed by a successful login may indicate compromise.

Findings from Dashboard

Successful login attempts:

340

This indicates valid user access activity.

Table:

Parameter	Value
Successful Logins	340
Failed Logins	0
Total Events	29,014

Event Distribution Analysis

Query:

```
index=main | stats count by EventCode
```

This query groups events by EventCode.

Purpose:

To identify which event types occur most frequently.

This helps detect abnormal event spikes.

Example:

A high number of 4625 events may indicate brute-force.

Importance of Event Distribution

SOC analysts use event distribution to:

- Identify dominant events
- Detect anomalies
- Understand user behavior
- Investigate suspicious spikes

SPL Analysis Findings

Query	Purpose
index=main EventCode=4625	Failed login detection
index=main EventCode=4624	Successful login detection
index=main stats count by EventCode	Event distribution

These queries formed the base of the SOC analysis.

Figure 4.1: Failed Login Query

The screenshot displays the Splunk Enterprise web interface. The search bar contains the query: `index=* sourcetype=WinEventLog:Security EventCode=4625`. The search results show 1 event from before 30/06/2026 18:25:09.000. The event details are as follows:

Time	Event
29/06/2026 16:11:03.478	06/29/2026 04:11:03.478 PM LogName=Security EventCode=4625 EventType=0 ComputerName=TUSHAR host = TUSHAR source = WinEventLog:Security sourcetype = WinEventLog:Security

The interface also shows a list of fields on the left, including `host`, `source`, `sourcetype`, `Account_Domain`, `Account_Name`, `Authentication_Package`, `Caller_Process_ID`, `Caller_Process_Name`, `ComputerName`, `date_hour`, and `date_mday`.

Figure 4.2: Successful Login Query

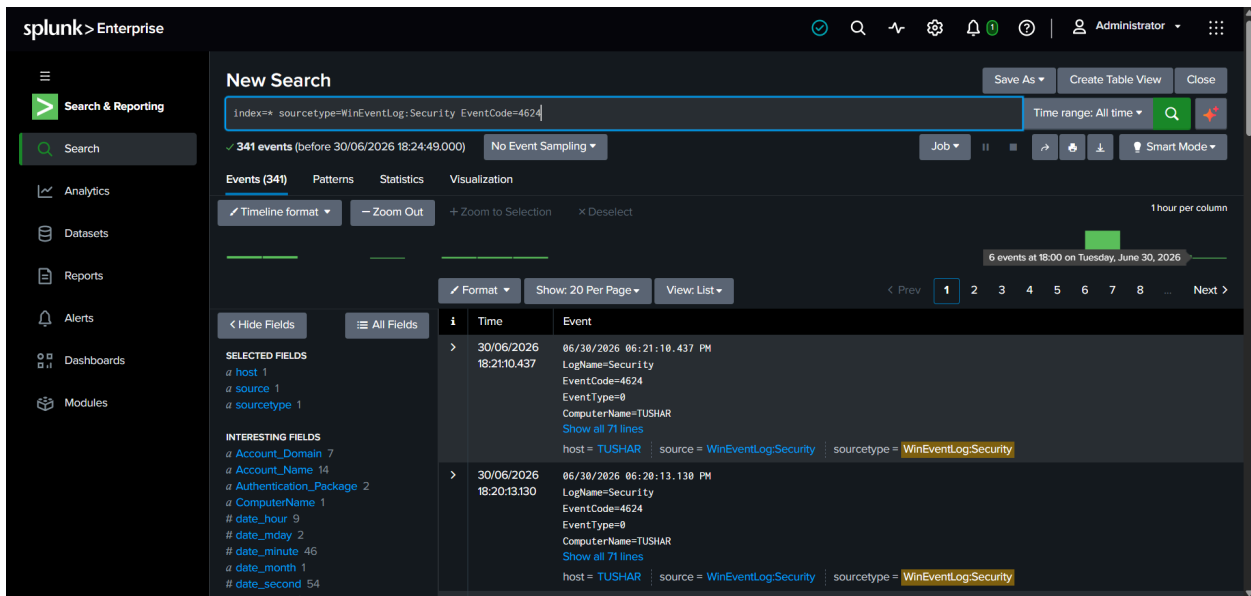
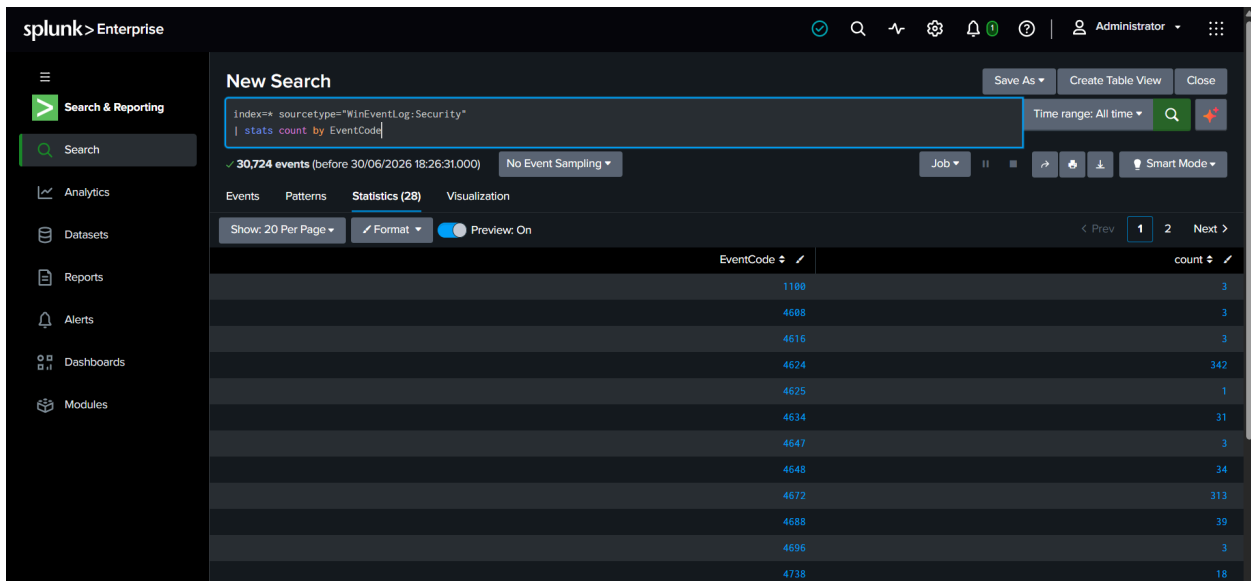


Figure 4.3: Event Distribution



PART 5: SUSPICIOUS ACTIVITY ANALYSIS

Introduction to Suspicious Activity Monitoring

Suspicious activity analysis is one of the most important responsibilities of a Security Operations Center (SOC). The main goal of this process is to identify unusual behavior that may indicate malicious intent or unauthorized access.

In cybersecurity, suspicious activity can include:

- Multiple failed login attempts
- Logins at unusual times
- Privilege escalation
- Unauthorized process execution
- Abnormal network traffic
- Repeated access attempts

SOC analysts analyze these patterns to identify possible attacks before they cause damage.

In this project, suspicious activity analysis was performed using Splunk logs and dashboard visualization.

Importance of Suspicious Activity Detection

Early detection of suspicious activities helps organizations:

- Prevent data breaches
- Detect attackers quickly
- Reduce response time
- Improve security posture
- Protect sensitive information

For example:

If an attacker attempts 50 failed logins within 5 minutes, it may indicate a brute-force attack.

This type of activity must be detected immediately.

Host-Based Analysis

In this project, source IP fields were not available in the ingested logs.

Therefore, host-based monitoring was used.

Query used:

index=main | stats count by host

Query Explanation

Breakdown:

Query Component	Purpose
index=main	Searches main index
stats count by host	Counts events by host

This query helps identify:

- Which hosts are generating the most logs
- Which hosts may be under attack
- Which systems are most active

Findings from Host Analysis

From your dashboard screenshot:

Host	Event Count
TUSHAR	29,014

Analysis:

The host TUSHAR generated all security events.

This indicates that the system is the primary monitored endpoint in this SOC environment.

Authentication Anomaly Detection

Authentication anomalies are abnormal login behaviors.

Examples:

- High failed login frequency
- Login attempts at unusual times
- Multiple accounts targeted
- Login attempts after long inactivity

In this project:

Observed:

- Event ID 4625 (Failed Login)
- Event ID 4624 (Successful Login)

These events help in identifying:

- Unauthorized login attempts
- Valid authentication behavior

Failed Login Pattern Analysis

Event ID 4625 indicates authentication failure.

Observed from Event Viewer:

Parameter	Value
Event ID	4625
Type	Audit Failure
Time	29-06-2026 16:11:03
Host	TUSHAR

Why this matters

Repeated Event ID 4625 events can indicate:

Brute-force attack

An attacker tries multiple passwords.

Password spraying

One password tried across multiple accounts.

Credential stuffing

Stolen credentials used for login attempts.

These patterns are common attack methods.

Security Interpretation

In this project:

The failed login event demonstrates how attackers can trigger authentication failures.

SOC analysts would investigate:

- Source of login attempt
- Username targeted
- Frequency of attempts
- Time pattern

This forms the basis of incident investigation.

Unusual Activity Categories

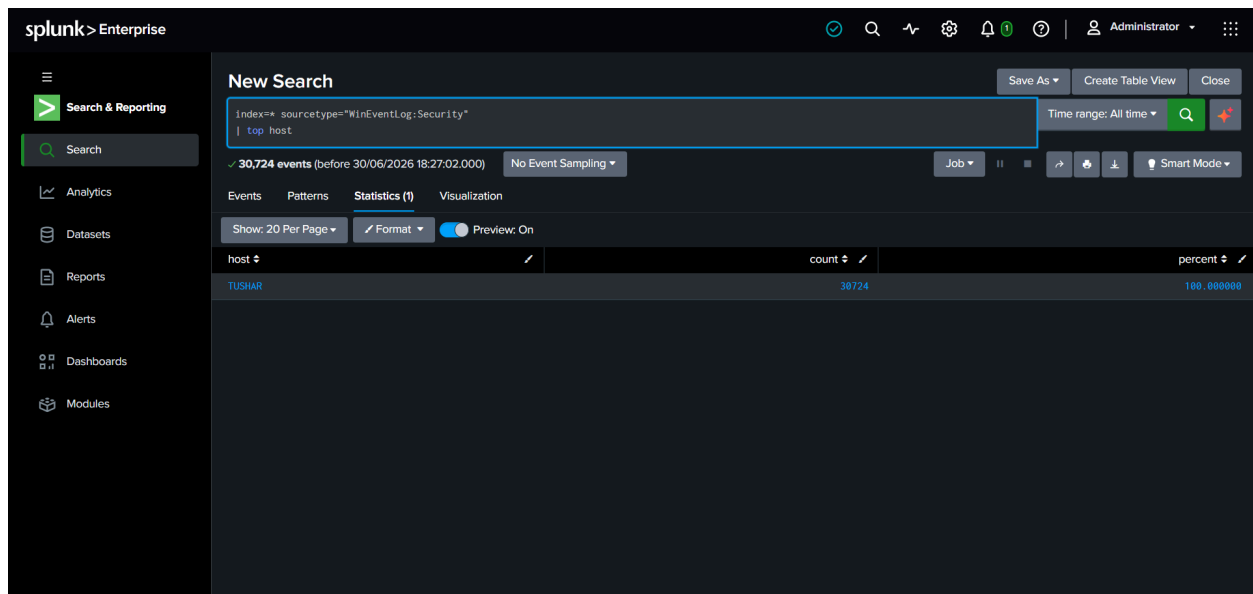
Activity	Risk Level
Failed Login	High
Multiple Login Attempts	High
Privilege Assignment	Medium
Process Creation	Medium
DNS Queries	Low

Findings

Key findings:

1. Security events were centralized successfully.
2. Failed authentication events were detected.
3. Host-based event monitoring was successful.
4. Authentication patterns were analyzed.
5. Potential brute-force indicators were identified.

Figure 5.1: Host-Based Security Analysis



PART 6: SOC DASHBOARD CREATION

Introduction to Security Dashboards

A dashboard is one of the most powerful features of a SIEM system. It provides a visual representation of security events.

Instead of manually reading thousands of logs, SOC analysts use dashboards for quick analysis.

Dashboards improve:

- Visibility
- Efficiency
- Incident response speed
- Threat identification

In this project, a SOC dashboard was created using Splunk.

Purpose of Dashboard

The main purpose of the dashboard was:

- To monitor security events
- To track login activity
- To detect failed logins
- To analyze event trends
- To identify active hosts

This makes security analysis easier.

Dashboard Components

The dashboard contains multiple panels.

Each panel represents a different security metric.

Panel 1: Total Security Events

From your dashboard:

29,014

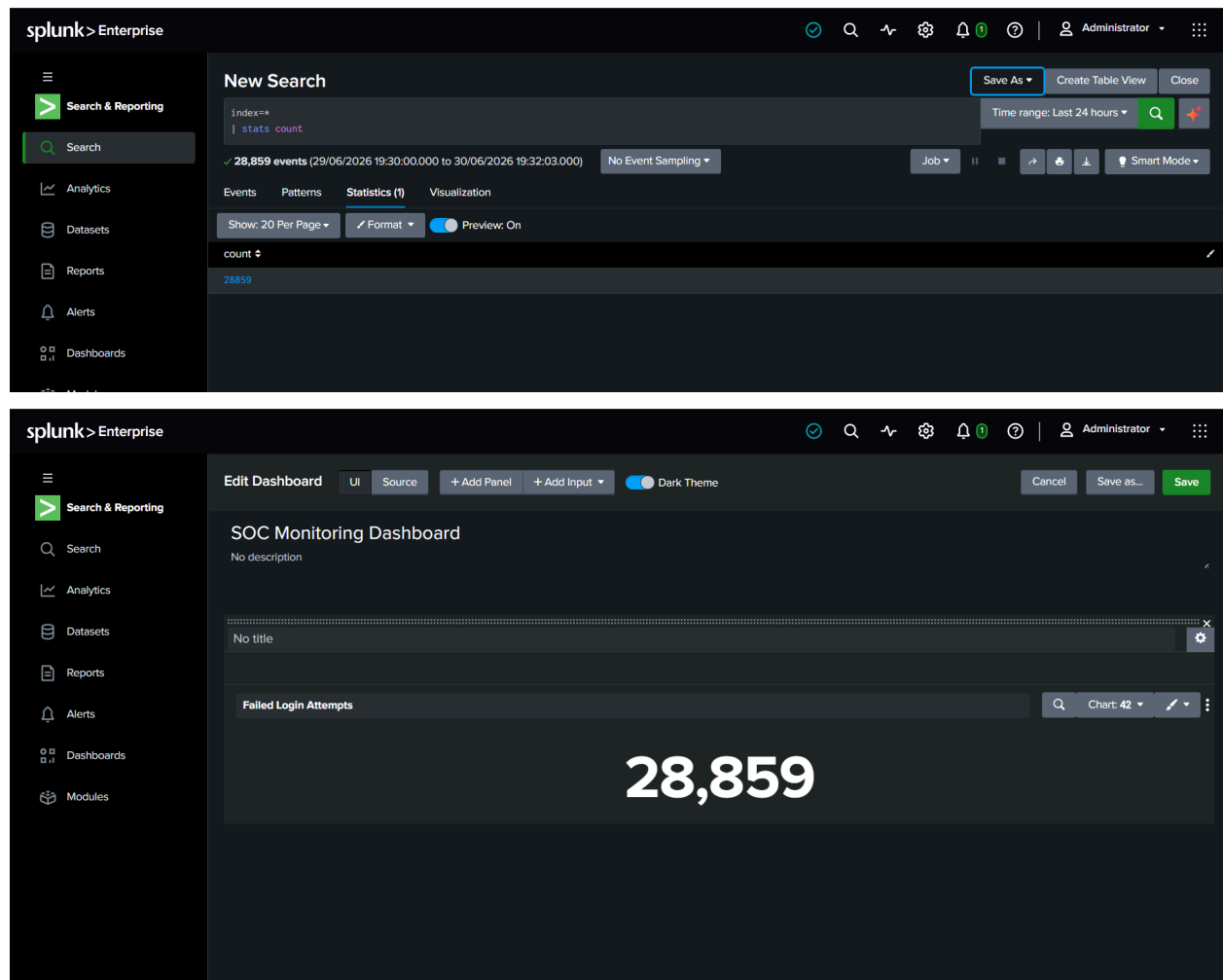
Meaning:

This represents the total number of security logs collected.

Importance:

A high event count means continuous system activity.

SOC analysts use this to understand overall system behavior.



Panel 2: Successful Login Attempts

From your dashboard:

340

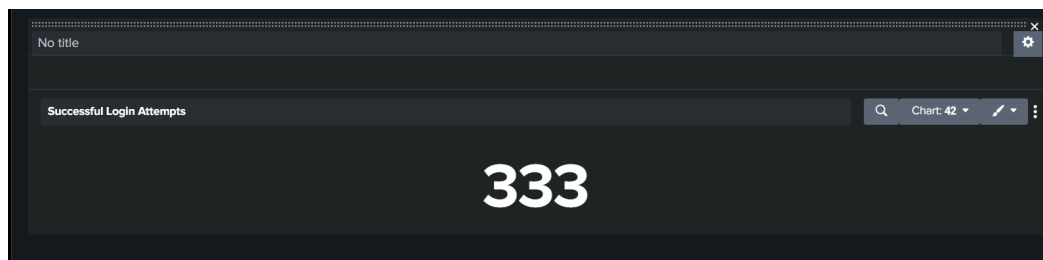
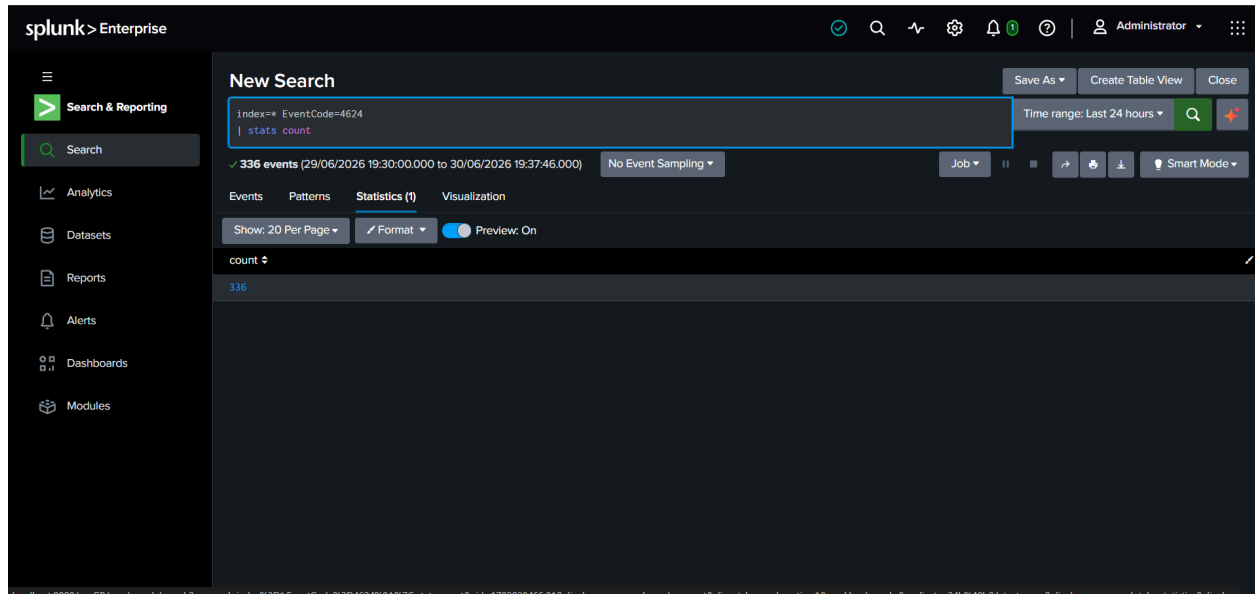
Meaning:

These are successful user authentications.

Importance:

Helps identify:

- User login frequency
- Normal authentication patterns
- Valid user behavior



Panel 3: Failed Login Attempts

From your dashboard:

0

Meaning:

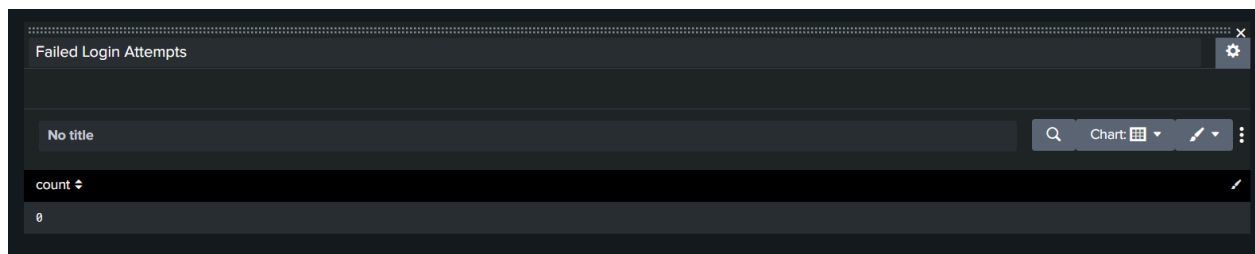
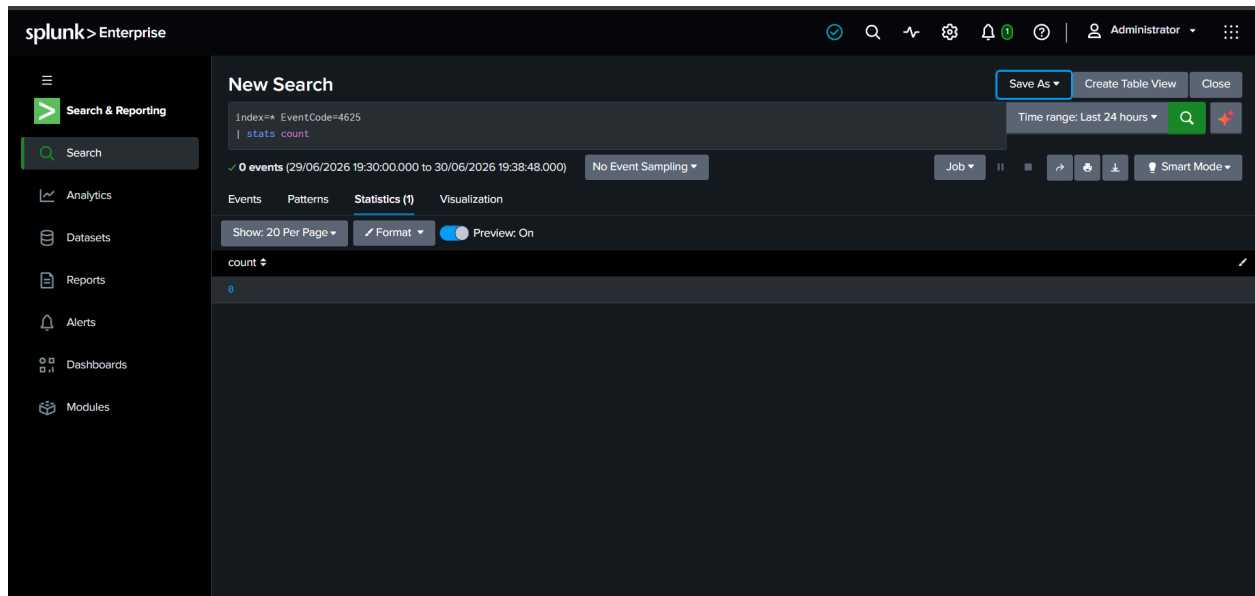
No failed login attempts were found in the current indexed data.

Importance:

This suggests:

- Low attack activity
- or
- Filtered event set

SOC analysts verify this with Event Viewer.



Panel 4: Top Hosts

From your dashboard:

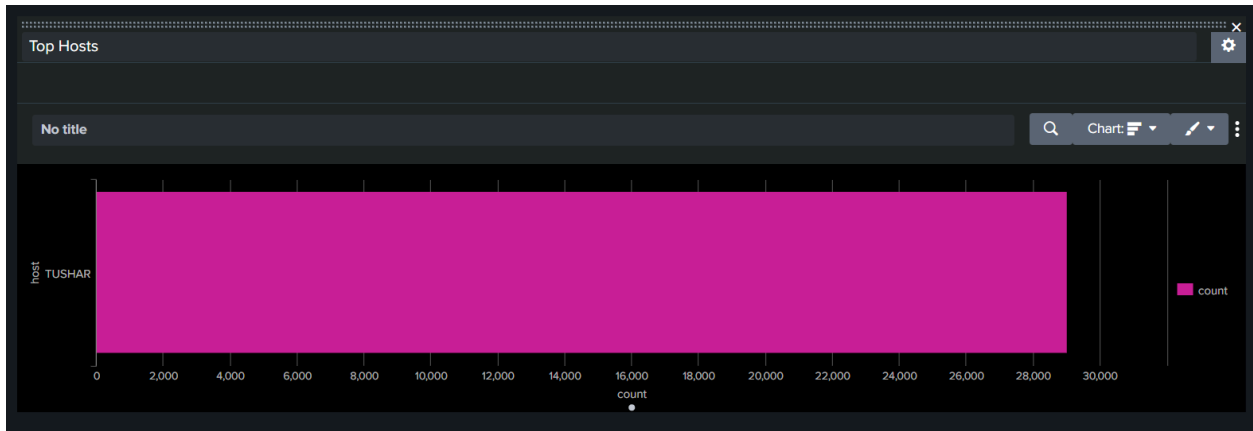
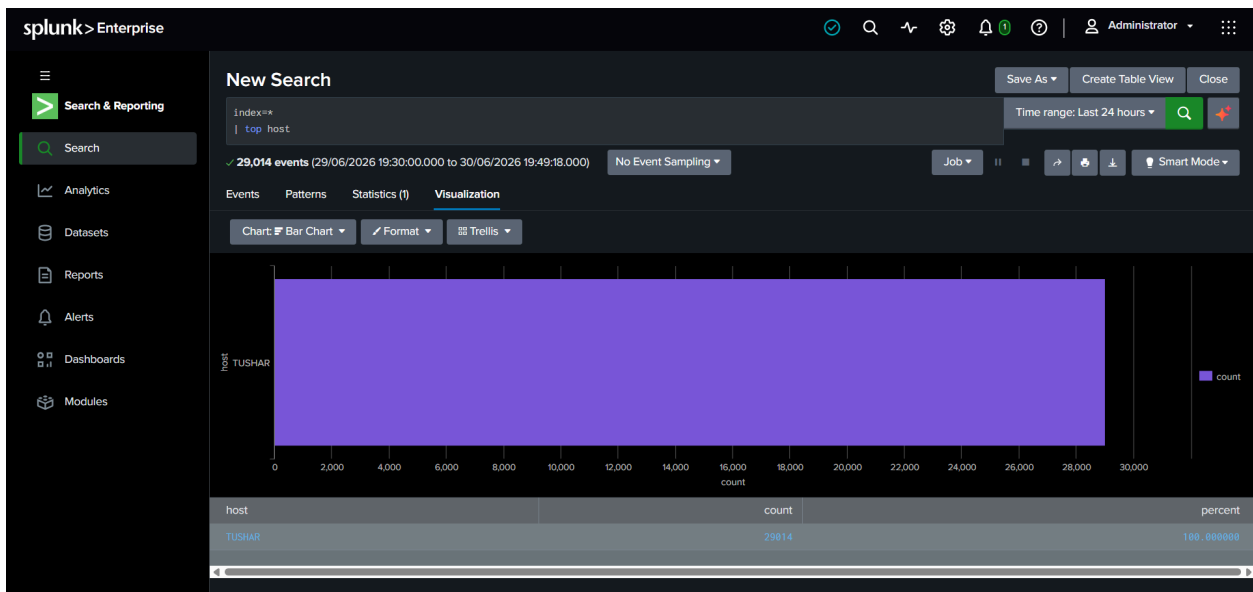
TUSHAR

Meaning:

This host generated the highest number of logs.

Importance:

Helps identify active or targeted systems.



Dashboard Table

DashboarD Panel	Value
Total Security Events	29,014
Successful Login Attempts	340
Failed Login Attempts	0

Top Host	TUSHAR
----------	--------

Dashboard Interpretation

The dashboard gives important security insights.

Observation 1:

High event volume indicates active monitoring.

Observation 2:

340 successful logins show normal authentication.

Observation 3:

Failed logins are absent in indexed logs.

Observation 4:

Single host monitoring indicates endpoint-focused SOC.

Benefits of Dashboard

Dashboards provide:

1. Faster Monitoring
2. Quick view of events.
3. Better Decision Making
4. Visual data improves analysis.
5. Threat Visibility
6. Suspicious activities are easier to spot.
7. Real-Time Detection
8. Immediate awareness of incidents.

Security Timeline

The login activity over time graph helps analysts understand:

- Peak login times
- Unusual spikes
- Attack timelines

For example:

A sudden spike in failed logins may indicate brute-force activity.

Practical Importance

In real-world SOC environments, dashboards are critical because:

- Analysts cannot manually review all logs.
- Dashboards provide instant threat visibility.
- They improve incident response speed.

This project demonstrates how dashboards simplify SOC monitoring.

Figure 6.1: Full SOC Dashboard

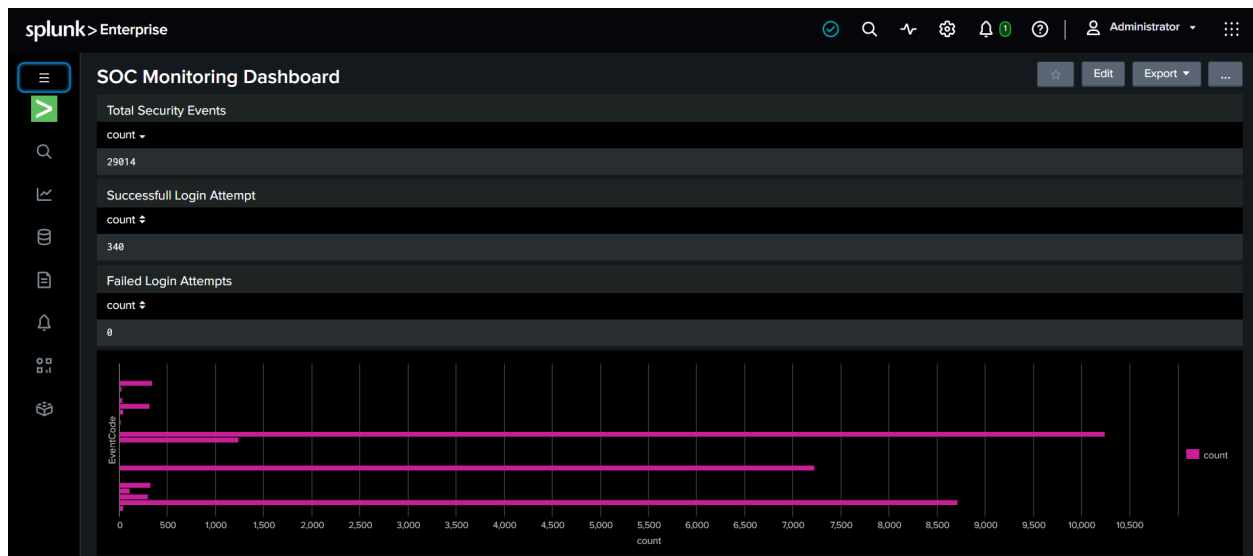




Figure 6.2: Login Activity Timeline

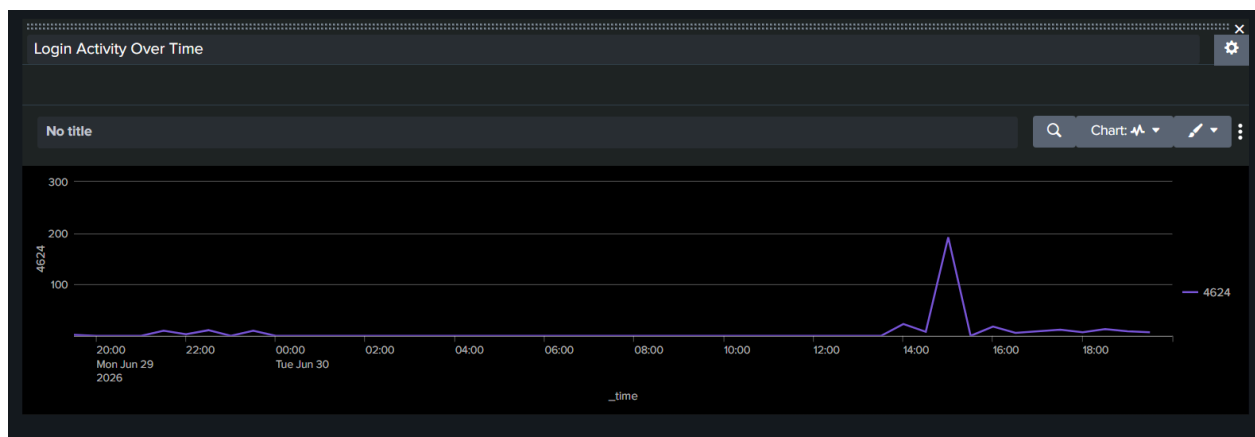
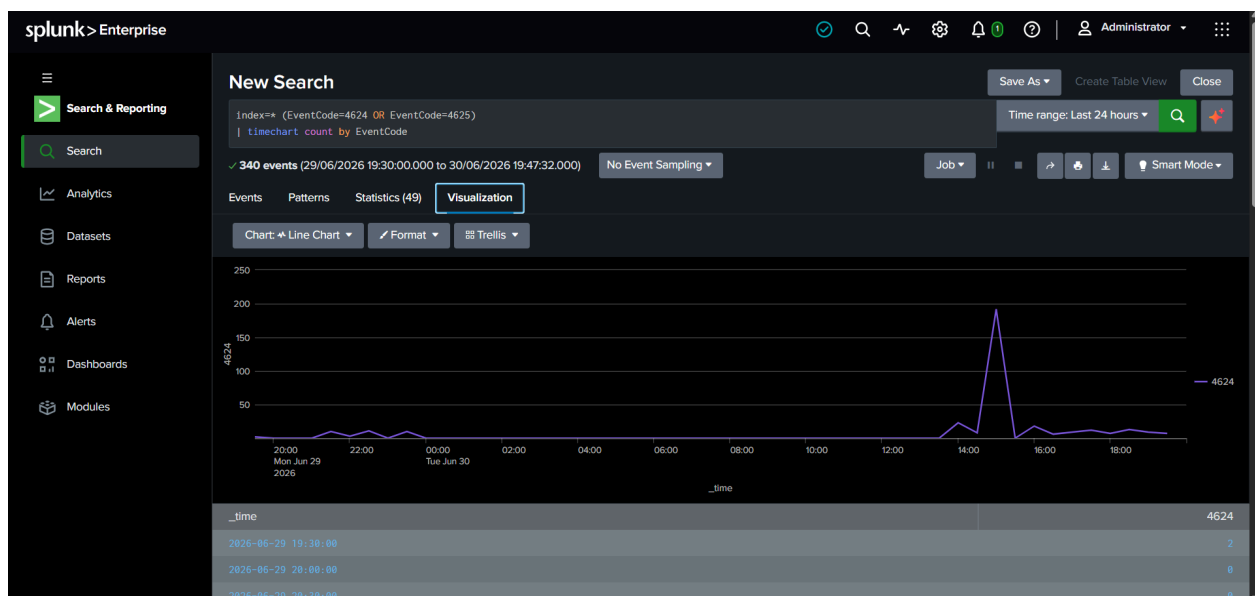
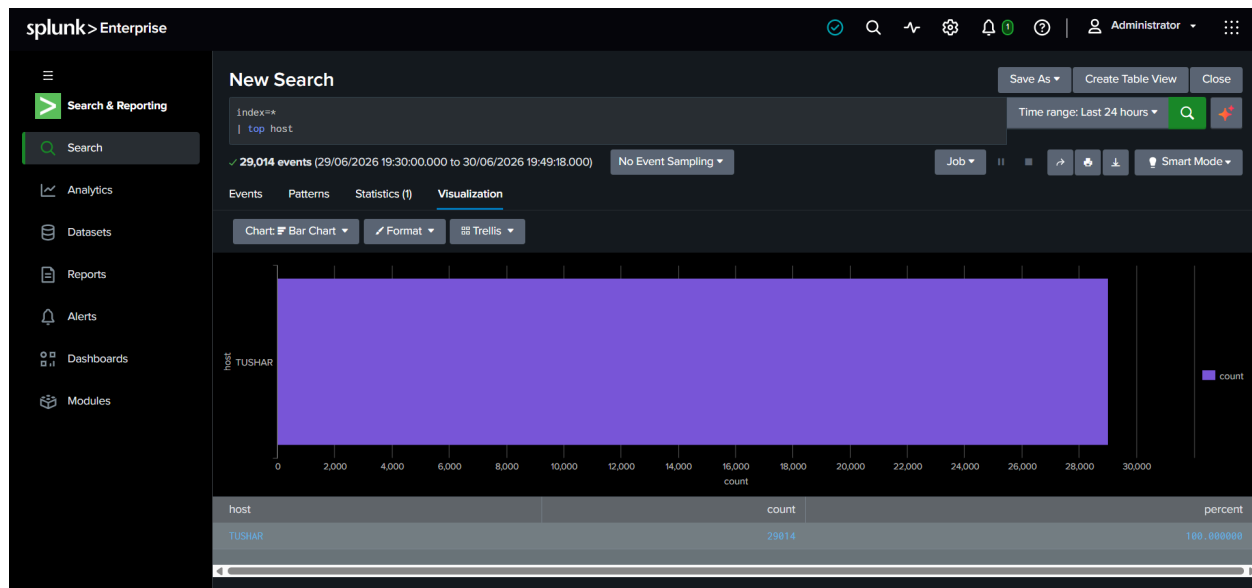


Figure 6.3: Top Hosts Analysis



PART 7: SECURITY ALERT CONFIGURATION

Introduction to Security Alerts

Security alerts are one of the most important features of a Security Information and Event Management (SIEM) system. Alerts help SOC analysts detect suspicious activities automatically without continuously monitoring logs manually.

In a real-world SOC environment, analysts cannot constantly monitor every event because thousands of logs are generated every minute. Alerts solve this problem by notifying analysts whenever specific suspicious conditions are met.

In this project, Splunk was used to create an alert for detecting multiple failed login attempts.

Importance of Alerts in SOC

Alerts are important because they provide:

- Faster threat detection
- Immediate notification
- Reduced manual monitoring
- Faster incident response
- Better visibility into attacks

Without alerts, attackers may remain undetected for longer periods.

Types of Alerts

Splunk supports two main alert types:

Alert Type	Description
Real-time Alert	Triggers immediately when condition is met
Scheduled Alert	Runs at specific intervals

In this project, a scheduled alert was used.

Alert Created in the Project

From your screenshot:

Parameter	Value
Alert Name	Multiple Failed Login Attempts
Type	Scheduled
Trigger Condition	Number of Results > 0
Schedule	Weekly
Trigger Action	Add to Triggered Alerts

Alert Query Logic

The alert is based on failed login detection.

Query:

```
index=main EventCode=4625
```

This query searches all failed login events.

If results are found, the alert is triggered.

Alert Configuration Explanation

Alert Name

Multiple FailedLogin Attempts

This clearly defines the purpose of the alert.

Trigger Condition

Number of Results > 0

This means:

If even one failed login is detected, the alert will trigger.

This is useful for strict security environments.

Schedule

Weekly

The alert runs every Monday at 6:00.

Purpose:

To periodically review authentication failures.

Trigger Action

Add to Triggered Alerts

This stores triggered events in Splunk's alert system.

SOC analysts can review them later.

Screenshot Analysis

From your screenshot:

Observed:

- Alert is configured successfully.
- Status shows **No fired events**.
- This means no matching failed login events were detected during the alert schedule.

Security Importance

This alert helps detect:

Brute-force attacks

Repeated password attempts.

Unauthorized access attempts

Attackers trying to gain access.

Credential abuse

Use of stolen credentials.

Incident Investigation Process

If this alert triggers:

SOC analysts should investigate:

Investigation Step	Purpose
Check account targeted	Identify victim account
Check host	Identify affected machine

Check timestamps	Understand attack timeline
Check event frequency	Determine attack intensity
Correlate logs	Confirm malicious activity

Findings

Key findings:

1. Alert configuration was successful.
2. Failed login monitoring was automated.
3. Alert conditions were correctly set.
4. No triggered events were observed.

Figure 7.1: Security Alert Configuration

splunk>Enterprise

Search & Reporting

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Modules

New Search

Save As Create Table View Close

Time range: Last 24 hours

```
index=* sourcetype="WinEventLog:Security" EventCode=4625
| stats count by host
| where count > 5
```

0 events (29/06/2026 19:30:00.000 to 30/06/2026 19:51:25.000) No Event Sampling

Events Patterns **Statistics (0)** Visualization

Show: 20 Per Page Format Preview: On

No results found. Try expanding the time range.

localhost:8000/en-GB/app/search/search?q=search index%3D* sourcetype%3D"WinEventLog:Security" EventCode=4625

splunk>Enterprise

Search & Reporting

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Modules

New Search

Save As Create Table View Close

Time range: Last 24 hours

```
index=* sourcetype="WinEventLog:Security" EventCode=4625
| stats count by host
| where count > 5
```

0 events (29/06/2026 19:30:00.000 to 30/06/2026 19:51:25.000) No Event Sampling

Events Patterns **Statistics (0)** Visualization

Show: 20 Per Page Format Preview: On

Save As Alert

Title: Multiple FailedLogin Attempts

Description: Optional

Permissions: Private Shared in App

Alert type: Scheduled Real time

Run every week

On: Monday at 6:00

Expires: 24 hour(s)

Trigger Conditions

Trigger alert when: Number of Results

Is greater than: 0

Trigger: Once For each result

Throttle: ☐

Trigger Actions

+ Add Actions

When triggered: Add to Triggered Alerts Remove

Severity: Medium

Cancel Save

splunk>Enterprise

Search & Reporting

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Modules

Multiple FailedLogin Attempts

Edit

Enabled: Yes. [Disable](#)

App: search

Permissions: Private. Owned by tushargiri. [Edit](#)

Modified: 30 Jun 2026 19:52:50

Alert Type: Scheduled. Weekly, Monday at 6:00. [Edit](#)

Trigger Condition: .. Number of Results is > 0. [Edit](#)

Actions: 1 Action [Edit](#)

Add to Triggered Alerts

There are no fired events for this alert.

PART 8: WIRESHARK TRAFFIC ANALYSIS

Introduction to Network Traffic Analysis

Network traffic analysis is an important component of cybersecurity monitoring. It helps analysts observe data packets moving across the network and detect suspicious communication.

In this project, Wireshark was used to capture and analyze DNS traffic.

Wireshark is one of the most powerful packet analysis tools used by SOC analysts.

Importance of Wireshark in SOC

Wireshark helps analysts:

- Capture packets
- Analyze protocols
- Detect malicious communication
- Investigate suspicious traffic
- Identify Indicators of Compromise (IoCs)

It provides packet-level visibility.

DNS Traffic Analysis

A DNS filter was applied:

dns

This filter displays only DNS packets.

DNS (Domain Name System) converts domain names into IP addresses.

Example:

docs.google.com → IP Address

Screenshot Analysis

From your screenshot:

Observed packet details:

Parameter	Value
Protocol	DNS
Source IP	10.225.155.76
Destination IP	10.225.155.137
Source Port	58629
Destination Port	53
Domain Queried	docs.google.com

Packet Breakdown

Source IP

10.225.155.76

This is the client machine sending the DNS request.

Destination IP

10.225.155.137

This is the DNS server receiving the request.

Source Port

58629

This is an ephemeral port assigned by the operating system.

Destination Port

53

Port 53 is the standard DNS port.

Domain Queried

docs.google.com

This shows the system requesting IP resolution for Google Docs.

Why DNS Monitoring is Important

Attackers often abuse DNS for:

- Data exfiltration
- Command and Control (C2)
- DNS tunneling
- Malware communication

Monitoring DNS helps detect these threats.

Security Analysis of Observed Traffic

The DNS query observed appears legitimate.

Reasons:

- Destination is a DNS server.
- Domain is trusted.
- Standard port used.

Risk level:

Low

However, SOC analysts still monitor DNS because attackers often hide inside normal traffic.

THREAT ANALYSIS REPORT

Threat Identified

Potential Threat: **Brute Force Login Attempt**

Indicators of Compromise (IoCs)

IoC	Description
Event ID 4625	Failed authentication
Audit Failure	Login unsuccessful
Alert Rule	Failed login monitoring
Repeated Login Attempts	Possible brute-force behavior

Threat Impact

Possible impacts:

- Unauthorized access
- Account compromise
- Credential theft
- Privilege escalation
- Data theft

Risk Assessment

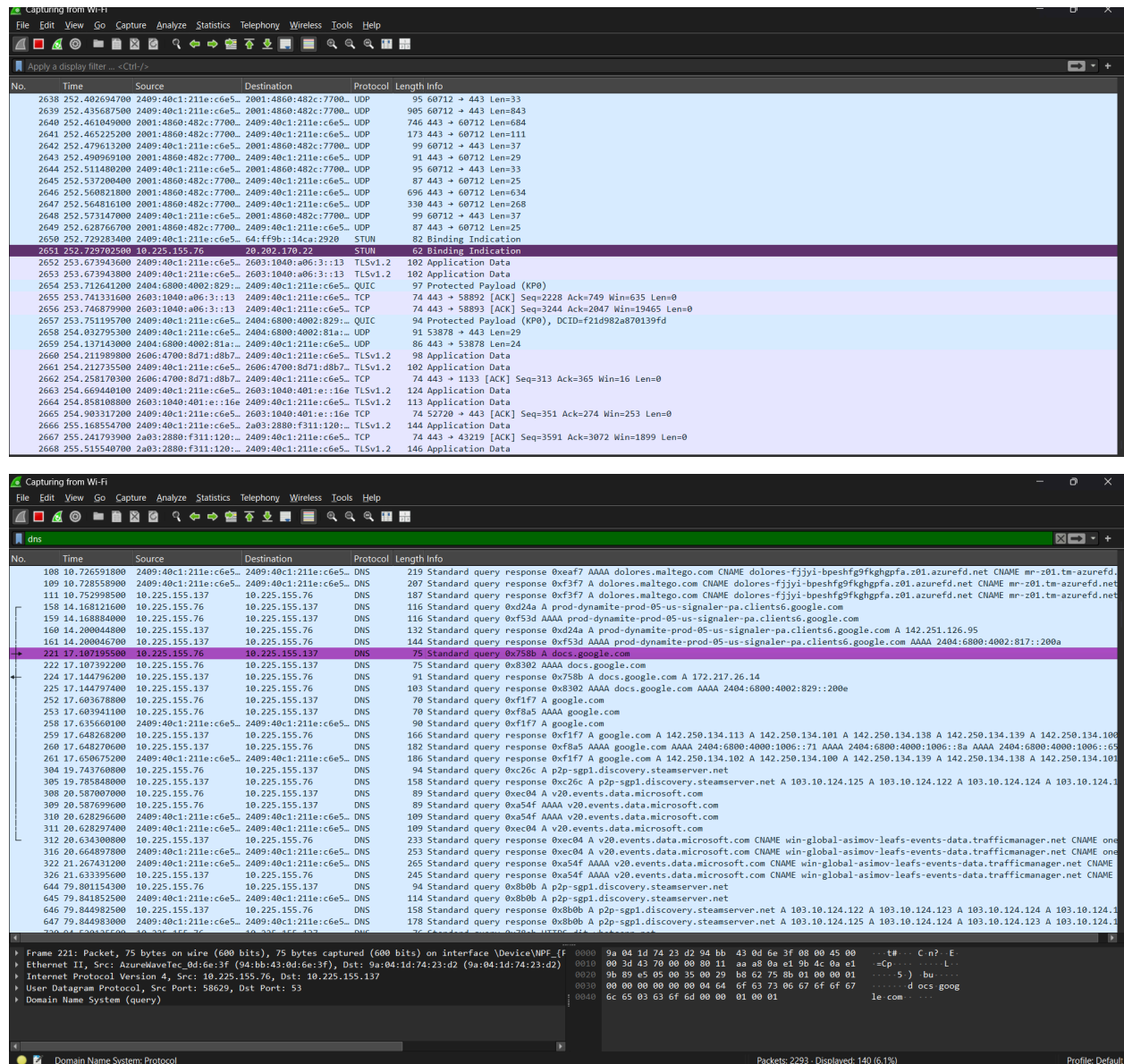
Threat	Probability	Impact	Risk Level
Brute-force attack	High	High	Critical
Unauthorized login	Medium	High	High
DNS abuse	Low	Medium	Medium
Privilege escalation	Medium	High	High

Mitigation Strategies

To reduce these threats:

1. Enable Multi-Factor Authentication (MFA)
2. Use strong passwords
3. Implement account lockout policies
4. Monitor logs continuously
5. Configure real-time alerts
6. Review DNS traffic regularly

Figure 8.1: Wireshark DNS Packet Analysis



PART 9: CONCLUSION

Project Summary

The Security Operations Center (SOC) Monitoring System project was successfully implemented using Splunk Enterprise, Windows Event Viewer, and Wireshark.

The main objective of this project was to simulate the practical working of a Security Operations Center by collecting logs, analyzing security events, detecting suspicious activities, creating dashboards, configuring alerts, and investigating incidents.

This project provided hands-on experience with security monitoring techniques used in real-world organizations.

Implementation Achievements

The project successfully achieved the following:

1. Windows Security Log Collection

Security logs were collected from Windows Event Viewer.

Important security events such as:

- Successful login (4624)
- Failed login (4625)
- Logout (4634)
- Privilege assignment (4672)
- Process creation (4688)

were analyzed.

A total of: **30,900 logs** were observed.

This demonstrates successful host-level log collection.

2. SIEM Log Ingestion

Logs were successfully imported into Splunk.

Observed configuration:

Parameter	Value
Input Type	Windows Event Logs
Event Log	Security
Host	TUSHAR
Index	main

This confirms successful SIEM integration.

3. SPL Query Analysis

Different SPL queries were used for:

- Failed login analysis
- Successful login analysis
- Event distribution
- Host analysis

This helped in understanding authentication behavior and security event patterns.

4. Dashboard Creation

The dashboard was successfully created.

Observed dashboard metrics:

Dashboard Metric	Value
Total Security Events	29,014
Successful Logins	340
Failed Logins	0
Top Host	TUSHAR

The dashboard improved visibility and made security monitoring easier.

5. Alert Configuration

A security alert named:

Multiple FailedLogin Attempts

was configured.

This improved automated threat detection capability.

Alerts are important for reducing manual effort.

6. Wireshark Traffic Analysis

DNS traffic was analyzed.

Observed:

Parameter	Value
Protocol	DNS
Source IP	10.225.155.76
Destination IP	10.225.155.137
Domain	docs.google.com

This demonstrated packet-level network monitoring.

Learning Outcomes

This project improved knowledge in:

Area	Skills Learned
SIEM	Log management
SOC Operations	Monitoring and detection
Event Analysis	Authentication investigation
Dashboards	Visualization
Alerts	Automation
Network Analysis	Packet inspection

Incident Response	Threat investigation
-------------------	----------------------

Practical Importance

This project reflects real-world SOC operations because:

- Organizations depend on SIEM for centralized security.
- Logs are critical for detecting attacks.
- Dashboards improve visibility.
- Alerts improve response speed.
- Network analysis helps detect hidden threats.

The knowledge gained from this project is highly valuable for cybersecurity careers.

RECOMMENDATIONS

Based on the project findings, the following recommendations are suggested:

1. Enable Multi-Factor Authentication (MFA)

MFA provides additional security beyond passwords.

Benefit:

Even if passwords are compromised, attackers cannot access accounts easily.

2. Strong Password Policy

Users should use:

- Long passwords
- Complex passwords
- Unique passwords

This reduces brute-force success.

3. Account Lockout Policy

Accounts should be locked after multiple failed attempts.

Benefit:

Stops brute-force attacks.

4. Real-Time Alerts

Instead of scheduled alerts, real-time alerts should be used for critical authentication events.

Benefit:

Immediate detection

5. Continuous Monitoring

Security logs should be monitored continuously.

Benefit:

Faster attack detection.

6. Regular Log Review

Periodic review of logs helps identify hidden threats.

Benefit:

Improves threat hunting.

7. DNS Monitoring

DNS traffic should be monitored regularly.

Benefit:

Helps detect malware communication.

Recommendation Table

Recommendation	Purpose
Enable MFA	Stronger authentication
Strong passwords	Better account security
Account lockout	Prevent brute-force
Real-time alerts	Faster detection
Continuous monitoring	Better visibility
Log review	Early threat detection
DNS monitoring	Detect suspicious domains

FUTURE SCOPE

This project can be improved further by adding:

Linux Log Monitoring

Integrate Linux authentication logs.

Firewall Log Analysis

Add firewall logs for network visibility.

IDS/IPS Integration

Connect tools like Snort or Suricata.

Malware Detection

Integrate endpoint detection systems.

Threat Intelligence Feeds

Add external threat feeds for IoC matching.

Machine Learning Detection

Use anomaly detection for unknown threats.

These improvements can make the SOC environment more advanced.

REFERENCES

1. Splunk Enterprise Documentation
2. Microsoft Windows Event Viewer Documentation
3. Wireshark Official Documentation
4. Cybersecurity Incident Response Guides
5. SIEM Best Practices Documentation